GMS-YOLO: A Lightweight Real-Time Object Detection Algorithm for Pedestrians and Vehicles Under Foggy Conditions

Yafei Chen, Yong Wang, Zheng Zou, and Wenxiu Dan

***Abstract*—In conditions of foggy weather, challenges, such as low light, blurred imagery, and dense fog that obscures target objects are prevalent. Moreover, computing resources are limited on edge devices. To tackle these challenges, a novel real-time detection algorithm GMS-YOLO for pedestrians and vehicles is proposed based on YOLOv10, which overcomes the semantic bottleneck of the model and enhances its detection performance. A novel ghost multiscale convolution (GMSConv) module is constructed, serving as the ghost multiScale feature extraction backbone network (GMS-Net). The Shape Consistent Intersection over Union (SCIoU) is introduced as the localization loss function, which takes into account the influence of the attributes of the regression box in the loss computation. Additionally, a compensatory consistency matching metric (CCMM) formula is designed to reduce the sensitivity of the original metric to IoU and regression scores. The GMS-YOLO algorithm has a lightweight structure, achieving FPS of 94 and 92 during the detection phase at the "'n"' and "'s"' sizes, respectively. Furthermore, we have deployed the model on Jetson Nano hardware, and the inference speed is also quite encouraging. We validated the effectiveness of the algorithm on the Foggy Cityscapes, RTTS, VOC2007-fog, and VOC2012-fog datasets. Experimental results indicate that GMS-YOLO outperforms the baseline model, with a mean average precision (mAP) improvement of 6.3% and 5.5% for the "n" and "s" scales, respectively. Consequently, the proposed GMS-YOLO algorithm not only demonstrates superior detection performance but also maintains a relatively low model complexity, significantly enhancing the efficiency and accuracy of object detection tasks in foggy environments. The source code for our algorithm is available at: https://github.com/Fwdchina/GMS-YOLO.**

***Index Terms*—Foggy object detection, ghost multiscale feature extraction backbone network (GMS-Net), lightweight structure,semantic bottleneck, shape consistent intersection over union (SCIoU).**

I. Introduction

CURRENTLY, autonomous driving technology has become one of the focal points of public interest. While autonomous driving systems can recognize most surrounding environments under normal conditions, their object detection performance is significantly compromised in foggy weather due to reduced visibility. In foggy conditions, the visibility of target objects is low, and fog often obscures these objects. This often results in suboptimal perception capabilities of the autonomous driving system in such environments [1], [2]. Additionally, most models designed for object detection tasks in foggy conditions are overly complex, making them unsuitable for practical applications due to the limited computational resources of edge devices. Effective foggy weather object detection technology can enhance the system's recognition under harsh weather conditions, ensuring that vehicles can accurately identify pedestrians and other vehicles [3], thus helping to avoid potential traffic accidents. Therefore, there is an urgent need for a more accurate and efficient object detection algorithm for foggy conditions.

In recent years, there have been significant advancements in the performance of object detection algorithms. Researchers have proposed various deep learning-based methods to address the challenge of object detection in foggy conditions, including two-stage detection methods [4], [5], [6], [7] and one-stage detection methods [8], [9], [10]. For two-stage detection methods, in [11], researchers designed Content-Parsing Generative Adversarial Networks (CP-GAN) to achieve the conversion between foggy and clear images. They then input these images into a detection model to mitigate the impact of foggy weather on the detection performance. Addressing issues like low visibility in fog, researchers in [12] proposed a sensor that enhances image visibility, where images are perceived and depth maps are generated through an image sensor in a stereo camera, followed by detection tasks using a joint detection algorithm. For one-stage methods, researchers in [13] and [14] developed an end-to-end foggy weather object detection algorithm by enhancing the model's feature modeling and representation capabilities.

For two-stage methods, although these methods have achieved good results, the introduction of prior knowledge or complex preprocessing dehazing networks leads to a larger computational load. This results in a more complex model. For one-stage methods, the detection performance is still not ideal. This article proposes a lightweight and efficient object detection algorithm for foggy weather based on YOLOv10, aimed at detecting pedestrians and vehicles under such conditions for autonomous driving.

The main innovations of this article are as follows.

1) Construct and utilize the ghost multiscale convolution (GMSConv) module to design ghost multiScale feature

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extraction backbone network (GMS-Net) as the backbone network of the GMS-YOLO algorithm, which enhances feature extraction capability and addresses the semantic bottleneck.
2) Use shape consistent intersection over union (SCIoU) Loss as the localization loss function renders the computation of loss more rational and improve its accuracy.
3) Adopt the Compensatory Consistent Matching Metric to improve the accuracy and robustness of model detection.

II. Related Work

A. Object Detection

Currently, there are two main frameworks for object detection: 1) the DETR (end-to-end object detection with transformer) [15], [16], [17] series based on transformers and 2) the you only look once (YOLO) [18], [19], [20], [21], [22], [23], [24], [25], [26], [27] series based on CNNs. Due to the distinctive architecture of transformers, they require longer training times and higher computational costs, which makes the DETR series less suitable for practical applications. In contrast, the YOLO series is known for its fast inference speed and accurate predictions, making it more suitable for real-world applications. Among the YOLO series algorithms, YOLOv10 [27] has achieved optimal performance. It removes the nonmaximum suppression (NMS) strategy used in YOLOv8 [25] and incorporates a series of lightweight modules. During training, both one-to-one and one-to-many dual-sign allocation strategies are employed, while only the one-to-one label assignment strategy is utilized during model inference, effectively reducing the computational load in the inference stage.

B. Localization Loss Function

The task of object detection involves classifying and localizing target objects within videos or images. The positioning is achieved through bounding box regression. A well-designed localization loss function can effectively accelerate the model's convergence speed and improve its accuracy. Yu et al. [28] proposed the concept of intersection over union (IoU), which measures the area of intersection between the ground truth box and the predicted box relative to the area of their union. However, this method does not account for boxe rotation. To address this limitation, Rezatofighi et al. [29] introduced generalized IoU (GIoU), which resolves the issue of identical overlap in the original IoU despite different box orientations. Building upon GIoU, Zheng et al. [30] proposed distance-IoU (DIoU), which minimizes the distance between two boxes, thereby significantly accelerating the model's convergence speed. Zhang et al. [31] introduced Efficient-IoU (EIoU), which breaks down the aspect ratio loss term into the difference between the predicted width and height and the dimensions of the minimum enclosing rectangle, further speeding up the model's convergence.

C. Object Detection in Foggy Weather

In foggy weather, haze can obscure target objects in images, resulting in suboptimal performance for detection algorithms. Based on this, researchers have proposed various deep learning-based methods, which can be broadly categorized into one-stage methods and two-stage methods.

For one-stage methods, the hazy image is directly input into the network for feature extraction, resulting in the final detection outputs. Meng et al. [8] introduced a target detection layer called swinfocus in YOLOv5s, which improves the model's ability to capture subtle features of target objects in hazy images. However, the added complexity of this approach significantly increases the computational burden, leading to slower inference speeds. Zhang and Hu [9] proposed a multiclass object detection framework capable of performing image enhancement, object classification, and localization tasks simultaneously. Despite this, under foggy conditions, the semantic information of images is obscured deep by the haze, and such enhancement methods remain limited. Zhang et al. [10] incorporated a high-resolution detection head into YOLOv5, which proves more effective for detecting small objects under foggy conditions. However, this approach is less suited for detecting vehicles and pedestrians in foggy weather. Wang et al. [32] developed an adversarial learning-based feature calibration module that effectively aligns feature representations across domains, from local to global, thereby reducing the distribution gap between normal and adverse weather conditions. Nonetheless, the distribution gap varies with weather severity, and under more severe weather conditions, the subtle features learned by this method remain constrained.

For the two-stage method, various prior knowledge is obtained by comparing the differences between the same image under normal weather and foggy conditions. This knowledge is then used to enhance the image, which is subsequently input into a detection network for the detection task. Liu et al. [4] proposed a differentiable image processing module that uses a standard 2-D convolutional network to predict image enhancement hyperparameters. Since the original image contains rich information while the predicted hyperparameters are limited, a significant amount of information is lost during the prediction process. Li et al. [5] introduced a domain-adaptive object detection framework, which performs adversarial information mining for difficult samples, improving the network's transfer learning capability and reducing the domain gap between image style and object appearance. However, learning domain gaps through adversarial networks requires expensive training time and a large number of samples, which limits the practicality and feasibility of the algorithm. Qin et al. [6] employed the Laplacian pyramid method to decompose the input image into low-frequency and multiple high-frequency components. The high-frequency components are then transformed using affine transformation to incorporate guidance information into the component features, while the low-frequency components are used to extract cross-level guidance information. However, when fine details or high-frequency noise are present in the image, it leads to blurred edges and loss of some information. Liu et al. [7] combined the GCANet [33] dehazing algorithm and a detection algorithm trained using edge and convolution feature fusion. Although edge extraction and

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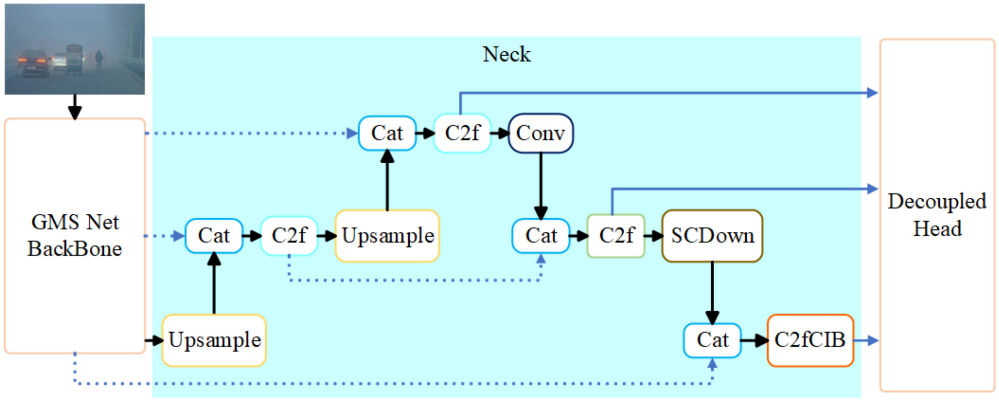


Fig. 1. GMS-YOLOn overall architecture, the black solid line represents data input to the next layer, the blue dashed line indicates skip connections, and the blue solid line represents output data to the decoupled detection head.

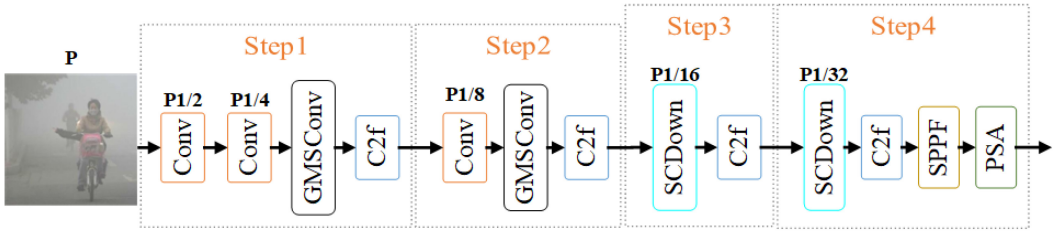


Fig. 2. Architecture of GMS-Net.

dehazing algorithms are well-developed, the method of fusing features using standard convolution still results in a loss of detailed semantic information. Since edge information is already scarce, this loss severely affects the effectiveness of information fusion. Shi and Song [34] used an enhanced U-Net framework for visual dehazing. The encoder and decoder use super-resolution back-projection to combine multilayer features and a back-projection feedback mechanism to improve image recovery, enabling the network to emphasize key channel and spatial information while reducing noise. However, the back-projection mechanism relies on the previous reconstruction result as the feedback input for the next round. When the image quality is poor, errors are constantly accumulated and amplified with each feedback, making image recovery difficult. Sen et al. [35] proposed a novel U-shaped residual dehazing module. Since the residual structure preserves some of the original semantic information of the hazy image, the dehazing effect is limited as a result. Qiu et al. [36] proposed a joint optimization dehazing module (AOD) based on the atmospheric scattering model and an image enhancement module (SAIP) to improve image quality. The AOD module uses standard 2-D convolution for defogging. Although it can remove some fog, it also results in the loss of original color information in the image, leading to partial distortion of the image.

III. Method

In foggy conditions, haze diminishes image contrast and distorts object shapes, impeding object detection algorithms from extracting deep semantic information. This limitation affects the model's performance, a situation we refer to as the "semantic bottleneck." Furthermore, fog's characteristics necessitate a detection model with robust feature extraction capabilities. To effectively tackle these challenges, we have developed the GMSConv object detection algorithm based on YOLOv10 (GMS-YOLO). The model's overall structure is depicted in Fig. 1, where "Conv" denotes a sequence comprising 2-D convolution, batch normalization, and SiLu activation. The architecture is divided into three main components. The backbone (GMS-Net), which is tasked with extracting features from the input image. Through downsampling, the channel count is increased, mapping the original image into a high-dimensional space to extract deep semantic information. The neck integrates feature maps of varying scales and depths using a feature pyramid method. The head (decoupled head), which conducts post-processing on three feature maps of different depths and sizes derived from the neck. Each feature map undergoes regression to yield predicted bounding box coordinates, confidence scores, and class predictions.

A. GMS-Net Design

We propose a hierarchical feature extraction backbone network, endowed with robust feature extraction capabilities, designed to surmount the semantic bottleneck. The fundamental component of the feature extraction submodule is the GMSConv. As depicted in Fig. 2, the structure of the GMS-Net is presented. The backbone network is segmented into four stages, with the assumption that the original image size is P (typically 640×640). In Step 1, the initial feature map is downsampled to 1/4 of its original size. This stage primarily carries out preliminary feature extraction, with a

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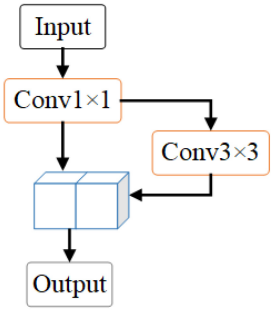


Fig. 3. Structure of GhostConv. The 1×1 and 3×3 convolutional kernels are employed. The number of channels is halved and then concatenated to restore the original channel count.

modest increase in the channel count. Steps 2 and 3 further downsample the feature maps to 1/8 and 1/16 of the original image size, respectively. During these stages, the image features are mapped into a high-dimensional space. Step 4 concludes the downsampling process by reducing the feature map to 1/32 of the original image size and employs a partial self-attention module (PSA) to bolster the backbone network's capacity for global representation learning. After carefully balancing the model's inference speed and accuracy, we integrated our proprietary GMSConv module in Steps 1 and 2. This integration not only enhances the model's feature extraction capabilities but also imposes a minimal additional computational load.

B. GMSConv Design

The GMSConv module is built upon GhostConv, which has demonstrated excellent performance in terms of high efficiency and low parameter count, as proven in [37]. Its structure is shown in Fig. 3. Here, considering that larger convolution kernels result in more parameters, we set the kernel sizes to 1×1 and 3×3. This combination allows the model to reduce the parameter count while still capturing subtle semantic information from the image. Additionally, it meets the requirements for feature extraction at the shallow layers of the model. The proposed GMSConv module is a novel lightweight structure. If the input tensor size is (1, 128, 160, 160), the module contains only 31168 parameters. Its structure is illustrated in Fig. 4. To manage the complexity of the module, we opt to use only two GhostConv blocks to extract features from semantic information at different depths. To avoid losing important semantic information, the outputs from the modules at different depths are concatenated and then fed into another GhostConv block for further fusion and extraction of deep semantic information. Finally, to ensure the module converges in the correct direction, the feature maps obtained after multiple extractions are added to the original input feature map of the module, resulting in the final output of the module.

C. SCIoU Loss Function

In object detection tasks, three types of loss functions are typically used for the joint optimization of the model: confidence loss, classification loss, and localization loss. In this

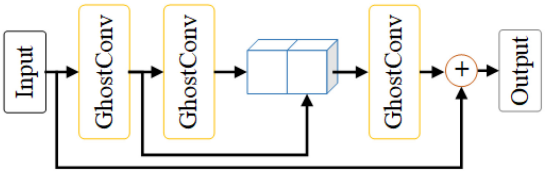


Fig. 4. Structure of GMSConv. GMSConv preserves raw features at different depths using concatenation and residual connections.

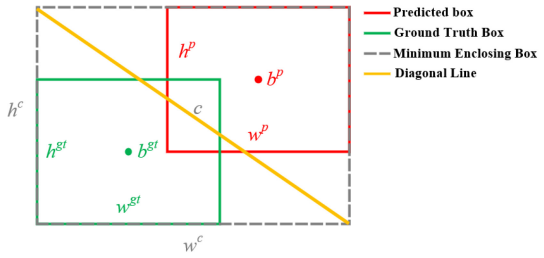


Fig. 5. Meaning of each symbol in SCIoU. The superscript "c" represents the attributes of the minimum enclosing box. The superscript "p" represents the attributes of the predicted box, and the superscript "gt" represents the attributes of the ground truth box.

article, we focus on the localization loss. Inspired by [38], we consider the impact of the attributes of the regression boxes on the IoU, leading to the introduction of the SCIoU localization loss. Its formula is as follows:

$$\mathrm{IoU} = \frac{\mathrm{area}(B_p \cap B_{gt})}{\mathrm{area}(B_p \cup B_{gt})} \tag{1}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w^p}{h^p}\right)^2 \tag{2}$$

$$\alpha = \frac{v}{(1 - IoU) + v} \tag{3}$$

$$\omega_w = 2 \times \frac{\left(w^{gt}\right)^s}{(w^{gt})^s + (h^{gt})^s} \tag{4}$$

$$\omega_h = 2 \times \frac{\left(h^{gt}\right)^s}{(w^{gt})^s + (h^{gt})^s} \tag{5}$$

$$\mathrm{Span} = \omega_h \times \frac{\left(x_c^p - x_c^{gt}\right)^2}{c^2} + \omega_w \times \frac{\left(y_c^p - y_c^{gt}\right)^2}{c^2} \tag{6}$$

$$\mathrm{ShapeL} = \sum_{i=w,h}\left(1 - e^{\gamma_i}\right)^{\varepsilon} \tag{7}$$

$$\begin{cases} \gamma_w = \omega_h \times \frac{|w - w^{gt}|}{\max(w, w^{gt})} \\ \gamma_h = \omega_w \times \frac{|h - h^{gt}|}{\max(h, h^{gt})} \end{cases} \tag{8}$$

$$L_{\mathrm{SCIoU}} = 1 - \mathrm{IoU} + \alpha v + \mathrm{Span} + \mu \cdot \mathrm{ShapeL}. \tag{9}$$

In (1), B_p and B_{gt} represent the predicted box and the ground truth box, respectively. In (2), w^{gt} and h^{gt} denote the width and height of the ground truth box, while w^p and h^p represent the width and height of the predicted box. The variable v indicates the difference in aspect ratio between the ground truth box and the predicted box. In (4) and (5), s is the size factor, which is related to the dataset. ω_w and ω_h are the weight coefficients for the horizontal and vertical directions, respectively. In (6), x_c^p and x_c^{gt} represent the x-coordinates of

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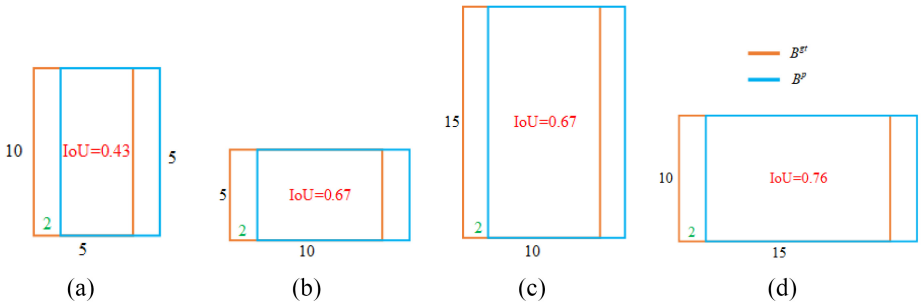


Fig. 6. Impact of bounding box size and offset direction on IoU with the same offset. (a) IoU = 0.43. (b) IoU = 0.67. (c) IoU = 0.67. (d) IoU = 0.76.

the center points of the predicted and ground truth boxes, while y_c^p and y_c^{gt} represent their y-coordinates. The variable c denotes the diagonal distance of the minimum enclosing rectangle between the predicted and ground truth boxes. In (7), ε is a hyperparameter. In (9), μ is a weight coefficient. The meanings of the various symbols are illustrated in Fig. 5.

SCIoU incorporates the properties of the boxes themselves, specifically their width and height, into the loss calculation, allowing the model to converge more accurately. As shown in Fig. 6, if we assume that the predicted box and the ground truth box are the same size, as depicted in (a) and (b), with a width and height of 5 and 10 respectively, and an offset of 2, different IoU values will be obtained when offsetting along the long side versus the short side, especially as the area becomes smaller. Furthermore, as illustrated in (a) and (c), even with the same offset size and direction, boxes of different sizes yield different IoU values. This situation is clearly unreasonable, which is why we introduce the horizontal and vertical weight coefficients from (4) and (5). These coefficients provide a more rational measure of the ratio of the distance between the centers of the predicted and ground truth boxes to the diagonal distance of their minimum enclosing rectangle.

D. Compensatory Consistent Matching Metric

In the model training phase, YOLOv10 adopts a dual-label assignment strategy, which includes both one-to-one and one-to-many label assignments. These two assignment methods use a consistency matching metric (CCMM) to quantitatively assess the level of consistency between them, enabling perceptual matching across the two branches. However, the original matching metric is computed by multiplying the spatial prior, the regression score, and the IoU of the detection boxes. This makes the metric score highly sensitive to the regression score and IoU. Specifically, when one of them is large and the other is small, the resulting metric score will also be small. This phenomenon leads to significant fluctuations in the metric score, which is detrimental to the training of one-to-one label assignment detection. Therefore, we propose a compensation-based CCMM, expressed by the following formula:

$$\text{ccmm}(\alpha,\beta) = s \cdot \left(\omega_1 \cdot p^{\alpha} \cdot \text{IoU}(B_p, B_{gt})^{\beta} + \omega_2 \cdot c(\alpha,\beta)\right) \quad (10)$$

$$c(\alpha,\beta)) = \text{IoU}\left(B_p, B_{gt}\right)^{\beta} + p^{\alpha}. \quad (11)$$

In (10), s represents the spatial prior [39], α and β are the metric coefficients [27], and ω_1 and ω_2 are the weight hyperparameters. In (11), $c(\alpha,\beta)$ is the compensation term of the metric formula. When the prediction performance is good, both the IoU and p approach 1, and the compensation term approaches 2. Due to the presence of the compensation term, even when there is a large discrepancy between IoU and p, the fluctuation of the metric score is reduced. This, in turn, decreases the sensitivity of the metric score to both factors, enabling better training of the one-to-one label assignment strategy and improving the model's accuracy.

TABLE I
EXPERIMENTAL CONFIGURATION PARAMETER TABLE

Configuration	Version Parameters
System	Ubuntu 22.04.4
GPU	NIVIDA RTX 2080 SUPER
Batch Size n/s	16/8
Max Epoch	500
Training Time of n/s (h)	4.3/8.3

IV. EXPERIMENTS

To validate the effectiveness of the algorithm, we conducted ablation experiments on three different foggy weather datasets, visualizing and comparing the feature maps from the original network with those processed by GMSConv, which demonstrates the powerful feature extraction capabilities of GMS-Net. Additionally, we compared the detection performance of the GMS-YOLOn/s algorithm and the YOLOv10n/s algorithm under foggy conditions. The experimental parameter configuration is shown in Table I. Due to device memory constraints, the batch size for the s-sized model is half that of the n-sized model. Our evaluation metrics and benchmark indicators were obtained on the same device, using the same default hyperparameters as the original YOLOv10. These hyperparameters include the optimizer, learning rate adjustment strategy, and the number of early stopping epochs, etc.

A. Datasets

We compared the model performance with multiple algorithms on the Foggy Cityscapes [40] and RTTS [41] datasets. Moreover, due to the limited availability of open-source datasets for object detection in foggy conditions, we constructed foggy versions of the VOC2007 [42] and VOC2012 [43] datasets using the fogging method from [10]

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TABLE II
DATASET CATEGORY COUNT TABLE

Dataset	Image	person	Car	Bus	Bicycle	Motorbike	rider	train	truck	Total
RTTS	4423	11536	25779	2638	710	1291	-	-	-	41954
VOC2007-fog	5011	5447	1644	272	418	390	-	-	-	8171
VOC2012-fog	17125	17401	2492	685	837	801	-	-	-	22216
Foggy Cityscapes	2965	17977	27048	382	3681	739	1768	171	489	52255

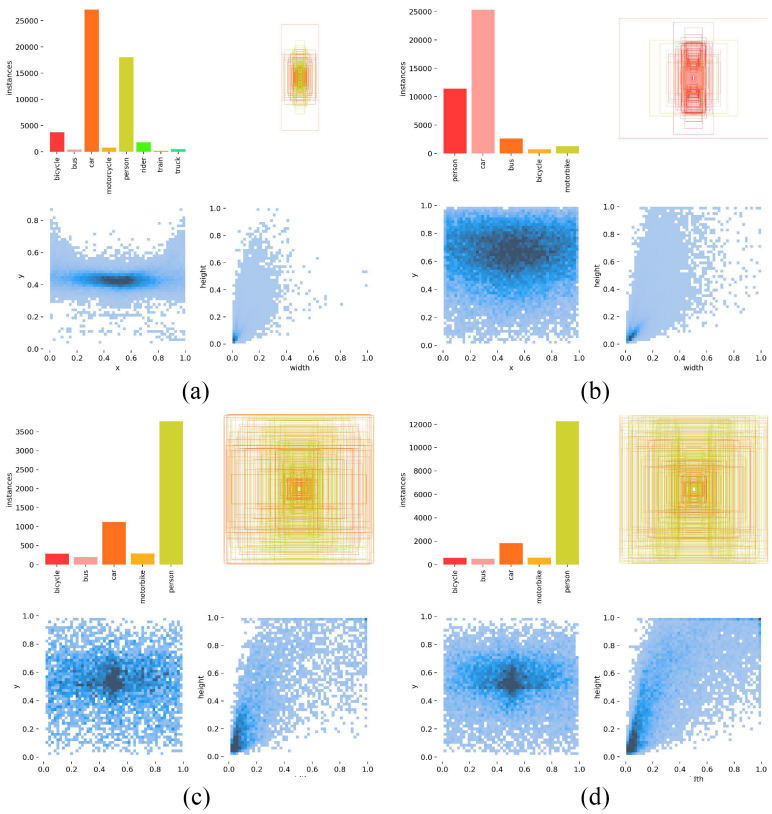


Fig. 7. Distribution of label counts and bounding box sizes, with (a), (b), (c), and (d) coming from the Foggy Cityscapes, RTTS, VOC2007-fog, and VOC2012-fog datasets, respectively.

to ensure a fair comparison. The labeling details of the four datasets are shown in Fig. 7.

For the VOC2007-fog and VOC2012-fog datasets, we first filtered the images and labels to retain only the five categories, consistent with the RTTS dataset, for our final dataset. We then combined all the images and randomly split the dataset into training and validation sets at a 7:3 ratio, with the number of samples in each category shown in Table II. Finally, we applied the atmospheric scattering model and the dark channel prior algorithm [44] to fog the dataset, with the formula represented as follows:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x)) \tag{12}$$

$$t(x) = e^{-\beta} \cdot d(x) \tag{13}$$

$$d(x) = s - 0.1 \cdot \rho. \tag{14}$$

In (12), $I(x)$ represents the foggy image, $J(x)$ denotes the clear image, $t(x)$ is the atmospheric transmission, and A is the atmospheric light value. The value of A is obtained by evaluating all images in the training set using the dark channel prior method from [10] and calculating the mean. The atmospheric transmission $t(x)$ can be computed using (13). Once $t(x)$ and A are known, the foggy image can be reconstructed from the clear image. For (13) and (14), β is the atmospheric scattering coefficient; s is the dimension of the fogging space; and ρ is the Euclidean distance from the current pixel to the center of the image. The fogging effect is illustrated in Fig. 8, where it is evident that the visibility of the fogged images is significantly low, making the target detection task quite challenging.

B. GMS-Net Feature Extraction

Due to the complexity of the GMSConv structure, and considering the balance between model inference speed and accuracy, we only added two GMSConv layers in the shallow layers of the YOLOv10 backbone to overcome the model's semantic bottleneck and enhance its feature extraction capability. These blocks are located at the 3rd and 6th layers of the backbone, as shown in Fig. 2. After training on the RTTS dataset, we computed the mean across all channels during

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TABLE III
COMPARISON OF ACCURACY AND MAP FOR VARIOUS CATEGORIES OF FOGGY DAY OBJECT DETECTION METHODS ON THE FOGGY CITYSCAPES DATASET. BOLD INDICATES THE BEST RESULTS

Model	bus	bicycle	car	mcycle	person	rider	train	truck	mAP
YOLOv8-n [25]	50.8	36.5	65.5	23.6	45.5	47.6	33.7	25.9	41.2
YOLOv8-s [25]	49.5	38.7	68.6	30.0	49.6	50.9	45.7	29.9	45.3
YOLOv10-n [27]	51.5	38.3	66.6	22.5	44.8	47.1	42.5	26.3	42.5
YOLOv10-s [27]	52.2	39.9	69.1	28.2	49.2	51.9	42.7	31.1	45.5
MTTrans [45]	45.9	46.5	65.2	32.6	47.4	33.8	49.9	25.8	43.4
DA-DETR [46]	45.8	46.3	63.1	31.6	49.9	50.0	37.5	24.0	43.5
PM-DETR [47]	47.2	46.1	64.7	32.4	47.8	39.6	**50.2**	26.5	44.3
CPLT [48]	42.5	**52.6**	67.8	**40.9**	**51.5**	**57.4**	30.5	27.9	46.4
SDA-DETR [49]	53.6	48.2	66.3	33.2	50.6	54.9	41.3	30.5	47.3
SR-DAYOLOv8 [50]	**53.9**	45.0	66.7	35.3	48.1	46.3	41.3	**37.6**	46.8
Ours-n	50.7	40.2	68.6	31.8	49.0	51.4	44.1	28.7	45.6
Ours-s	52.9	44.0	**69.3**	32.7	50.1	52.3	49.5	37.4	**48.5**

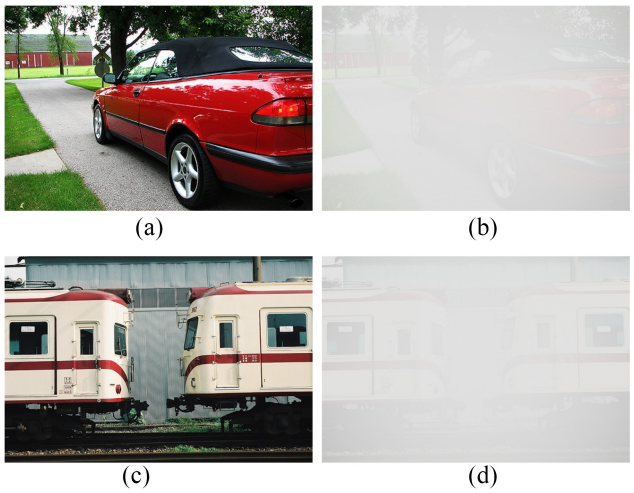


Fig. 8. Demonstration of fogging effects, with (a) and (c) being the original images from different datasets, and (b) and (d) being the fogged images of both. (a) VOC2007 Image. (b) VOC2007 Foggy Image. (c) VOC2012 Image. (d) VOC2012 Foggy Image.

the inference phase and visualized the heatmaps, as shown in Fig. 9.

For the shallow layers of the model, the heatmap regions in (b) and (c) are quite similar, indicating that their feature extraction capabilities are comparable. However, because the GMSConv undergoes multiple convolution operations, the boundary of the heatmap focus in (c) is less distinct compared to (b), resulting in a more visually blurred image. Comparing (d) and (e), it is clear that the target objects in (e) have a greater distinction from the background in terms of heatmap intensity, indicating that the GMSConv demonstrates better performance in the deeper layers of the model. The GMSConv effectively differentiates the target objects from the image background, significantly reducing the occurrence of missed detections. Therefore, it possesses superior feature extraction capabilities, allowing it to overcome the model's semantic bottleneck.

C. Method Testing and Verification

We use YOLOv10 as the baseline model, with evaluation metrics, including P (precision), R (recall), mAP50 (mean average precision (mAP) at IoU of 0.5), and FPS. The calculation formulas are presented as follows:

$$P = \frac{TP}{TP + FP} \tag{15}$$

$$R = \frac{TP}{TP + FN} \tag{16}$$

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n}\int_{0}^{1} P(R_i)d(R_i). \tag{17}$$

In (15) and (16), *TP* (true positive) indicates that a positive sample is correctly identified as a positive sample. *FP* (false positive) indicates that a negative sample is incorrectly identified as a positive sample. *FN* (false negative) indicates that a positive sample is incorrectly identified as a negative sample. n represents the number of classes in the dataset, which is 5 in this article, and FPS denotes the number of images processed per second.

1) Comparative Experiment: We chose to conduct comparative experiments on the Foggy Cityscapes and RTTS datasets because both datasets contain a large number of instances, feature smaller targets, and pose more challenges for detection, making them more representative. For comparison algorithms, we selected the most advanced algorithms currently available from the YOLO series, as well as some target detection algorithms specifically designed for foggy conditions. To ensure the real-time performance of the algorithms, we compared the two models with the smallest parameters: YOLOv8 [25], and the baseline model YOLOv10 [27]. Regarding the algorithms for foggy conditions, we selected MTTrans [45], DA-DETR [46], PM-DETR [47], CPLT [48], SDA-DETR [49], SR-DAYOLOv8 [50], RestorNet-YOLOX-s [51], TogetherNet [52], and RDMNet [53] for comparison, with the results shown in Table III and Table IV.

For Table III, the metrics for MTTrans, DA-DETR, PM-DETR, and CPLT are from article [49]. Compared to the baseline for the n size, our accuracy has improved for all classes except for the bus class, with the mcycle class showing a notable increase of 9.3%. For the s size, the accuracy for all classes has improved, and our mAP is the highest, reaching 48.5%.

For Table IV, the metrics for RestorNet-YOLOX-s and TogetherNet come from [53]. We generally consider algorithms to have real-time capability if the FPS during the

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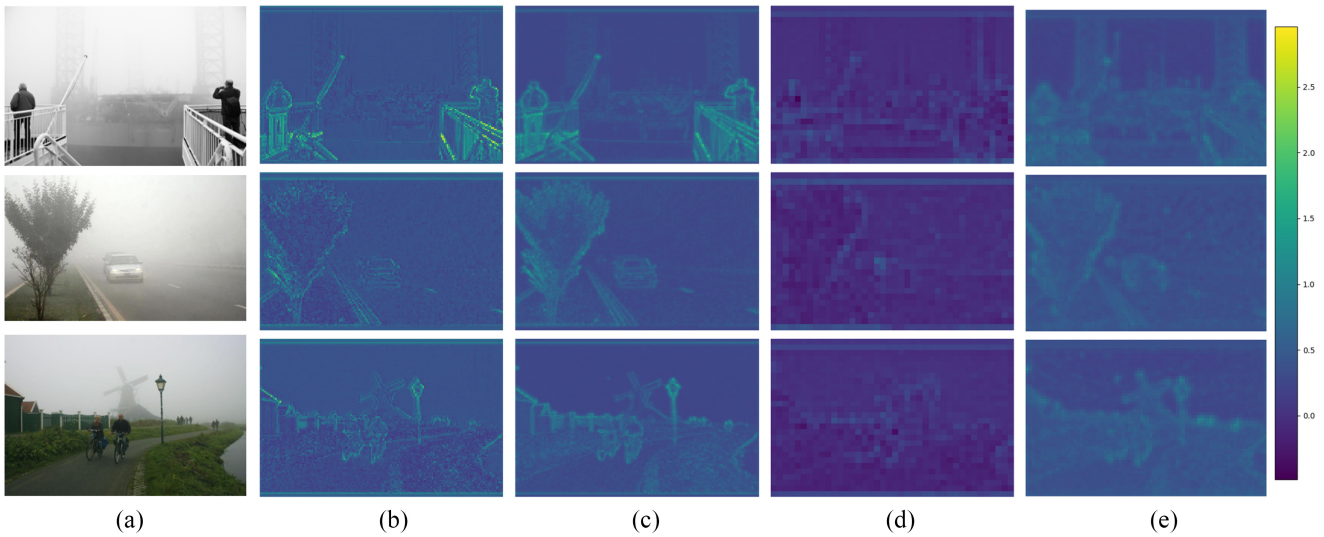


Fig. 9. Heatmap comparison: (a) is the original image, (b) and (d) are the heatmaps extracted from the 3rd and 6th layers of YOLOv10n, while (c) and (e) are the heatmaps from the same layers processed by GMS-Net.

TABLE IV
PERFORMANCE AND INFERENCE SPEED COMPARISON COMPARISON OF VARIOUS ALGORITHMS ON THE RTTS DATASET, WITH BOLD INDICATING THE BEST RESULTS

Model	$Parameters$	$GFLOPs$(G)	P(%)	R(%)	$mAP50$(%)	FPS
YOLOv8-n [25]	3006623	**8.1**	68.3	54.8	58.5	**135**
YOLOv8-s [25]	11127519	28.4	72.8	50.9	59.3	128
YOLOv10-n [27]	2686366	8.2	67.0	48.3	55.2	102
YOLOv10-s [27]	8038830	24.5	70.2	51.7	60.1	98
RestorNet-YOLOX-s [51]	-	182.6	-	-	52.4	-
TogetherNet [52]	-	30.9	-	-	57.5	-
RDMNet [53]	-	27.1	-	-	58.6	-
Ours-n	2861422	11.6	70.0	**57.5**	61.5	94
Ours-s	8229486	28.4	**73.1**	57.2	**65.6**	92

inference phase is greater than 35. It can be observed that all the algorithms we tested meet this criterion for real-time performance. At the same dimensions of "n" and "s", our accuracy, recall, and mAP are higher than the baseline by 3%/3.1%, 9.2%/5.5%, and 6.3%/5.5%, respectively. Furthermore, our recall rates are the first and second highest, with only a slight decrease in FPS. In other words, we achieved a significant performance improvement at the cost of a slight reduction in inference speed.

In addition, we compared our model with the baseline model on the VOC2007-fog and VOC2012-fog datasets. As shown in Fig. 8, the visibility of the fogged images is relatively low, making detection more challenging. Although the VOC2007-fog dataset contains over 5,000 images, the number of instances for each category is limited, with fewer than 10 000 instances in total, which is detrimental to model training. This leads to only modest performance improvements for our model on this dataset. As indicated in Table V, the mAP of the model under size "s" only increased by 0.3%, and by 1.0% under size "n". For the multiinstance dataset VOC2012-fog, our model still demonstrated considerable performance, with an increase in mAP of 0.7% and 1.4% compared to the baseline model for sizes "n" and "s", respectively.

Finally, we exported the trained model to the ONNX format and deployed it on edge devices for comparison.

TABLE V
PERFORMANCE COMPARISON TABLE OF GMS-YOLO AND BASELINE ALGORITHMS ON SYNTHETIC DATASETS

Dataset	Model	$Size$	P(%)	R(%)	mAP(%)
VOC2007-fog	Baseline	n	**77.5**	48.3	55.6
	Ours	n	77.3	**49.0**	**56.6**
	Baseline	s	74.0	**53.4**	59.7
	Ours	s	**79.8**	51.9	**60.0**
VOC2012-fog	Baseline	n	72.6	55.9	61.1
	Ours	n	**73.2**	55.9	**61.8**
	Baseline	s	68.2	**61.0**	63.3
	Ours	s	**70.5**	60.1	**64.7**

TABLE VI
FPS COMPARISON TABLE ON JETSON NANO

Model	$Size$	FPS
Baseline	n	3.8
GMS-YOLO	n	3.8
Baseline	s	3.4
GMS-YOLO	s	3.3

The device we selected is the Jetson Orin Nano8G, which is equipped with an NVIDIA Ampere architecture GPU featuring 32 Tensor Cores and 1024 cores, with a computing

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TABLE VII
ABLATION EXPERIMENT RESULTS TABLE

Size	Model	GMS-Net	CCMM	SCIoU	$Param$(M)	$FLOPs$(G)	$mAP^{val}(\%)$
n	BaseLine				2.69	8.2	55.2
	M1	✓			2.86	11.6	59.9
	M2		✓		2.69	8.2	60.3
	M3			✓	2.69	8.2	57.4
	M4	✓	✓		2.86	11.6	61.3
	Ours	✓	✓	✓	2.86	11.61	61.5
s	BaseLine				8.03	24.5	60.1
	M5	✓			8.23	28.4	62.8
	M6		✓		8.03	24.5	61.6
	M7			✓	8.03	24.5	62.0
	M8	✓	✓		8.23	28.4	63.8
	Ours	✓	✓	✓	8.23	28.4	65.6

TABLE VIII
ABLATION STUDY TABLE FOR SIZE FACTORS

Foggy Cityscapes		RTTS		VOC2007-fog		VOC2012-fog	
s	$mAP50(\%)$	s	$mAP50(\%)$	s	$mAP50(\%)$	s	$mAP50(\%)$
0.4	44.6/47.8	0.5	58.1/61.9	0.2	56.3/59.9	0.3	61.5/63.7
0.5	**45.6/48.5**	**0.6**	**61.5/65.6**	**0.3**	**56.6/60.0**	**0.4**	**61.8/64.7**
0.6	44.3/46.5	0.7	59.5/65.2	0.4	55.9/58.9	0.5	60.6/64.5

power of up to 40 TOPS. The images we used to test the inference speed were real-time images captured by the edge device's camera, and the results are shown in Table VI. It can be observed that the inference speed of our algorithm on this device is comparable to that of the baseline model, which fully demonstrates the practicality of the method we proposed.

2) Ablation Experiment: We conducted ablation experiments on several improvement methods proposed in this article to observe their impact on the performance of the baseline model and to demonstrate their effectiveness. The experimental results are shown in Table VII. Additionally, since the size factor of SCIoU needs to be adjusted based on the size of the label boxes in different datasets, we tested Foggy Cityscapes, RTTS, VOC2007-fog, and VOC2012-fog to determine their optimal values, with the test results presented in Table VIII. In addition, to further validate the effectiveness of SCIoU, we compared it with the original ShapeIoU [54], the baseline method BaseLine's CIoU [30], and the existing ShapeCIoU [54], and the results are shown in Table IX. Finally, we conducted an ablation experiment on the weight hyperparameters of the compensatory CCMM, and the results are shown in Table X.

In the M1 and M5 configurations, we use GMS-Net as the backbone network of the model. Experimental results show that this network performs well. For models of sizes "n" and "s", the mAP improves by 4.7% and 1.7%, respectively. In the M2 and M6 configurations, we introduce a compensatory CCMM on top of the baseline. This method improves the model with a one-to-one label assignment strategy to undergo more appropriate training. Experimental results demonstrate that after introducing the compensatory CCMM, the mAP for models of sizes "n" and "s" improves by 5.1% and 1.5%, respectively, achieving significant results. In the M3 and M7 configurations, we incorporate the SCIoU method into the baseline, making the localization loss calculation more reasonable and improving model accuracy. Experimental results show that this method improves the mAP for models of sizes "n" and "s" by 2.2% and 1.9%, respectively, yielding promising results. Furthermore, in the M4 and M8 configurations, we test both GMS-Net and the compensatory consistency matching method together. The results indicate that combining the two methods leads to an improvement over the individual experiments,

TABLE IX
ABLATION STUDY OF THE LOCALIZATION LOSS FUNCTION, WITH THE BOLD FONT INDICATING THE BEST RESULT

Model Size	Loss	$mAP50(\%)$
n	CIoU [30]	55.2
	ShapeIoU [38]	56.3
	ShapeCIoU [54]	54.6
	SCIoU(our)	**57.4**
s	CIoU [30]	60.1
	ShapeIoU [38]	60.3
	ShapeCIoU [54]	61.7
	SCIoU(our)	**62.0**

TABLE X
ABLATION EXPERIMENT ON THE COMPENSATION CCMM WEIGHT HYPERPARAMETERS BASED ON THE RTTS DATASET. THE BOLDED FONT INDICATES THE BEST RESULTS

Model Size	ω_1	ω_2	$mAP50(\%)$
n	0.8	0.1	60.4
	0.6	0.2	**61.5**
	0.4	0.3	59.3
s	0.8	0.1	63.8
	0.6	0.2	**65.6**
	0.4	0.3	63.9

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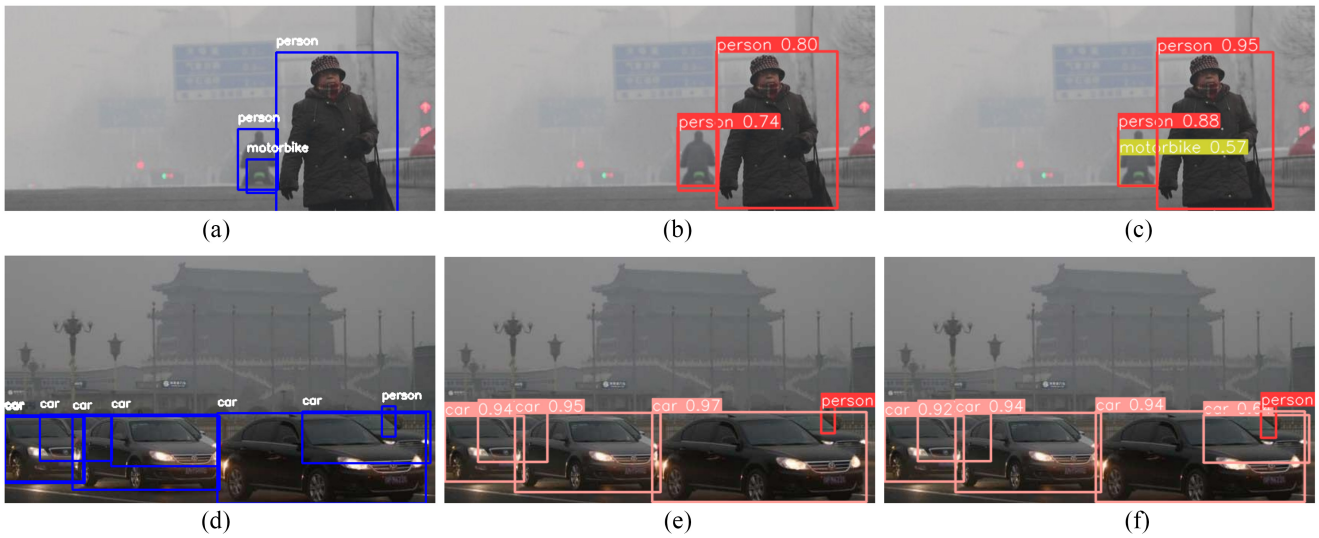


Fig. 10. Comparison of Detection Results for the Model at Size "n": (a) and (d) are the ground truth labels, (b) and (e) are the results from the baseline model, and (c) and (f) are the results from GMS-YOLO.

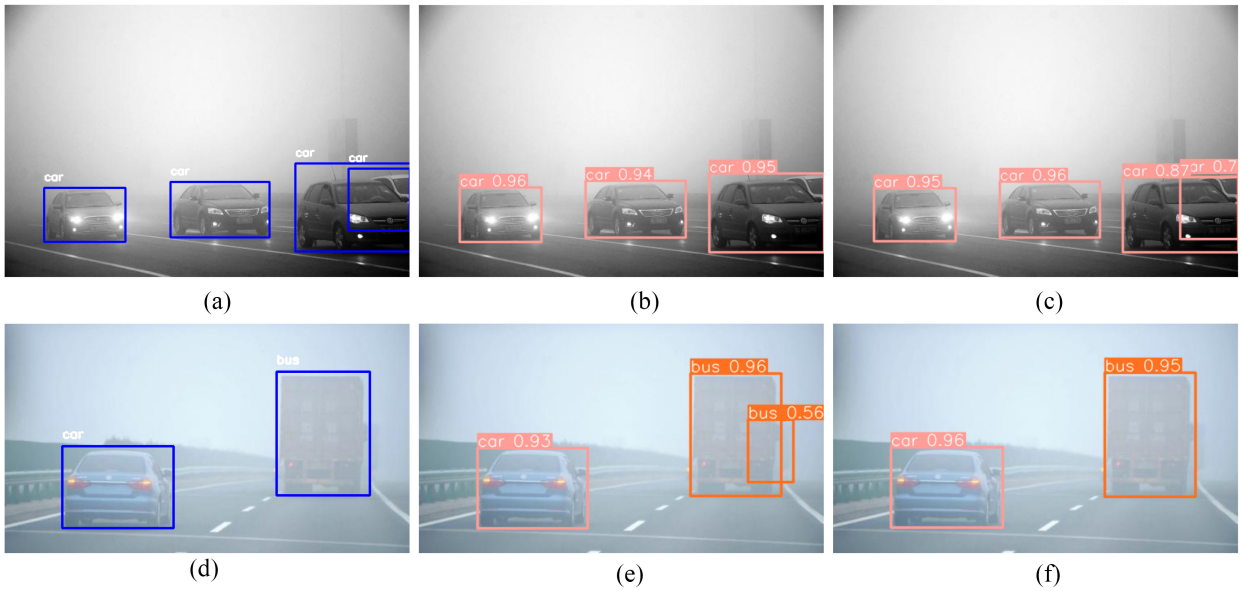


Fig. 11. Comparison of Detection Results for the Model at Size s: (a) and (d) are the ground truth labels, (b) and (e) are the results from the baseline model, and (c) and (f) are the results from GMS-YOLO.

suggesting that these methods complement each other well. Compared to the baseline, the mAP of the two models improves by 6.1% and 2.7%, respectively. Finally, when comparing our GMS-YOLO with the baseline model, the mAP for models of sizes "n" and "s" increases by 6.3% and 5.5%, respectively. Although the parameter count increases by 6.3% and 2.5%, and the model complexity increases by 41.5% and 15.9%, the GMS-YOLO with size "n" has 66.4% and 52.7% lower parameter count and model complexity, respectively, compared to the "s" size baseline, while achieving a 1.4% increase in mAP.

Moreover, due to the varying sizes of bounding boxes across different datasets, larger bounding box areas correspond to smaller values for the size factor. Therefore, multiple tests are needed to determine the optimal value. In this section, we evaluated the performance of GMS-YOLO with different size factor values at sizes "n" and "s", with the results shown in Table VIII. For the Foggy Cityscape, RTTS, VOC2007-fog, and VOC2012-fog datasets, the optimal size factors were found to be 0.5, 0.6, 0.3, and 0.4, respectively. As shown in Fig. 7, the target boxes in the RTTS dataset tend to be generally smaller, which is why its size factor is larger than those for the other two datasets a finding that is also supported by Table VIII. Furthermore, we further validated the effectiveness of SCIoU. Under the original YOLOv10 model, we compared the original ShapeIoU, the baseline localization loss CIoU, and the existing improvement method ShapeCIoU with our proposed method, and the results are shown in Table IX. Since ShapeCIoU proposed in [54] is designed for small object detection tasks, its performance under foggy conditions may

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be limited. The results demonstrate that, whether compared with the two original methods or the existing ShapeCIoU, our proposed SCIoU outperforms all of them.

Finally, to identify the appropriate weight hyperparameters for the compensatory CCMM, we evaluated the performance of two models with different sizes, *n* and *s*, on the RTTS dataset. As shown in Table X, our experiments revealed that the optimal combination of weight hyperparameters is 0.6 and 0.2.

3) Detection Results: Figs. 10 and 11 show visual examples of detection results for the GMS-YOLO algorithm and the baseline YOLOv10 algorithm on the RTTS dataset at sizes "n" and "s", respectively. Our method can effectively detect objects with subtle features, such as the motorbike in (c) of Fig. 10 and the car in the lower right corner of (f). Additionally, our method can accurately identify target objects, reducing false positives. For instance, in (e) of Fig. 11, the baseline model incorrectly detects a bus on the right side, while our method accurately identifies it, as shown in (f) of Fig. 11. The comparison demonstrates that our method is more capable of detecting local features and distinguishing target objects from the background, effectively reducing missed detections and overcoming the semantic bottleneck of the model.

V. Conclusion

To address the issues of blurred boundaries between target objects and the environmental background, unclear images, and low light conditions in foggy weather, we propose a GMS-YOLO target detection algorithm based on the YOLOv10 framework. We constructed the GMSConv using GhostConv to build the GMS-Net backbone network, enhancing the model's feature extraction capabilities and overcoming the semantic bottleneck. To improve accuracy, we introduced the SCIoU localization loss, which takes into account the intrinsic properties of the target boxes in the localization loss calculation, making it more reasonable. Additionally, we proposed a compensatory CCMM based on the original CCMM method. This approach reduces the sensitivity of the measurement scores to IoU and regression scores, allowing for better training of detection performance under one-to-one label strategies, thereby improving model accuracy. We conducted comparative experiments with multiple algorithms on the synthetic dataset Foggy Cityscapes and the publicly available real-world dataset RTTS. Due to the limited availability of public foggy datasets, we synthesized our own dataset using the fogging method based on VOC2007 and VOC2012 to validate the effectiveness of our algorithm. Experimental results show that GMS-YOLO demonstrates strong feature extraction capabilities under foggy conditions, effectively overcoming the semantic bottleneck of the model and enhancing its accuracy.

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